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Substrate treating apparatus and substrate transfer robot

Abstract

The present invention provides a substrate treating apparatus. The substrate treating apparatus includes: a liquid treatment chamber configured to treat a substrate with a liquid; a drying chamber configured to dry the liquid-treated substrate; a transfer robot configured to transfer the substrate between the liquid treatment chamber and the drying chamber, and including a hand which is movable along an X-axis, a Y-axis, and a Z-axis and is rotatably driven based on the Z-axis, and on which the substrate is placed; an optical system configured to photograph a form of a liquid film of the substrate, in which when the substrate is transferred from the liquid treatment chamber to the drying chamber, the substrate is wetted with a chemical liquid and is transferred by the transfer robot in a state of being formed with a liquid film formed; and a controller configured to measure the form of the liquid film photographed by the optical system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2021-0063629 filed in the Korean Intellectual Property Office on May 17, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a transfer robot for transferring a substrate and a substrate treating apparatus including the same.

BACKGROUND ART

(3) In order to manufacture a semiconductor device, a desired pattern is formed on a substrate through various processes, such as photography, etching, ashing, ion implantation, and thin film deposition. Various treatment liquids are used in each process, and contaminants and particles are generated during the process. In order to solve this problem, a cleaning process for cleaning contaminants and particles is performed before and after each process.

(4) In general, the cleaning process performs a liquid treatment process and a dry treatment process. In the liquid treatment process, a treatment liquid is supplied on the substrate, and in the dry treatment process, the liquid remaining on the substrate is removed. The liquid treatment process includes a chemical treatment step, a rinse operation, and a solvent replacement operation. The chemical treatment operation removes foreign substances on the substrate with chemicals, the rinse operation removes chemicals remaining on the substrate, and the solvent replacement operation replaces the rinse solution remaining on the substrate with an organic solvent.

(5) This will be described with reference to FIG. 1. In general, the chemical treatment operation, the rinsing operation, and the solvent replacement operation are performed in a liquid treatment chamber **260**, and the drying treatment operation is performed in a drying chamber **280**.

Accordingly, the substrate on which the solvent replacement operation has been completed is transferred to the drying chamber **280**.

(6) Semiconductor equipment for cleaning substrates usually has an n-column and m-layer structure. In this case, the liquid treatment chamber **260** and the drying chamber **280** are randomly matched rather than matched with each other on the same layer. If the substrate processed in the liquid treatment chamber **260** of the first-row and first-floor is matched with the second-row and third-layer drying chamber, the substrate moves in the X, Y, and Z-axes and rotates, so that the behavior of the solvent is the most severe.

SUMMARY OF THE INVENTION

(7) The present invention has been made in an effort to provide a substrate treating apparatus capable of efficiently treating a substrate and a substrate transfer robot.

(8) The present invention has also been made in an effort to provide a substrate treating apparatus capable of solving drying defects that may occur due to a process in which a substrate moves between a liquid treatment chamber and a drying chamber, and a substrate transfer robot.

(9) The present invention has also been made in an effort to provide a substrate treating apparatus and a substrate transfer robot, which are capable of solving a problem from a part unit when the problem occurs after drying completion by providing a monitoring system for a liquid film formed by being discharged on a substrate of a substrate W in a substrate treatment facility including a liquid treatment chamber.

(10) The object of the present invention is not limited thereto, and other objects not mentioned will be clearly understood by those of ordinary skill in the art from the following description.

(11) An exemplary embodiment of the present invention provides a substrate treating apparatus, including: a liquid treatment chamber configured to treat a substrate with a liquid; a drying

chamber configured to dry the liquid-treated substrate; a transfer robot configured to transfer the substrate between the liquid treatment chamber and the drying chamber, and including a hand which is movable along an X-axis, a Y-axis, and a Z-axis and is rotatably driven based on the Z-axis, and on which the substrate is placed; an optical system configured to photograph a form of a liquid film of the substrate, in which when the substrate is transferred from the liquid treatment chamber to the drying chamber, the substrate is wetted with a chemical liquid and is transferred by the transfer robot in a state of being formed with a liquid film formed; and a controller configured to measure the form of the liquid film photographed by the optical system.

(12) In the exemplary embodiment, the optical system may transmit first data obtained by photographing the form of the liquid film to the controller in the state where the transfer robot picks up the substrate in the liquid treatment chamber, before the transfer robot injects the substrate into the drying chamber, the optical system may transmit second data obtained by photographing the form of the liquid film to the controller, and the controller may compare the first data and the second data, and when the form of the liquid film of the second data is out of an error range in comparison with the first data, the controller may determine whether to load the substrate into the drying chamber and stores the second data.

(13) In the exemplary embodiment, the optical system may be a camera which photographs an edge region of the substrate from a lateral surface of the substrate.

(14) In the exemplary embodiment, the camera may be provided to photograph a first lateral surface of the substrate and a second lateral surface that is the other side of the first lateral surface.

(15) In the exemplary embodiment, the controller may measure a first contact angle that is a contact angle of the chemical liquid with respect to the first lateral surface of the substrate and a second contact angle that is a contact angle of the chemical liquid with respect to the second lateral surface of the substrate.

(16) In the exemplary embodiment, the optical system may transmit first data obtained by photographing the form of the liquid film to the controller in the state where the transfer robot picks up the substrate in the liquid treatment chamber, before the transfer robot injects the substrate into the drying chamber, the optical system may transmit second data obtained by photographing the form of the liquid film to the controller, the controller may measure a first contact angle that is a contact angle of the chemical liquid with respect to the first lateral surface of the substrate and a second contact angle that is a contact angle of the chemical liquid with respect to the second lateral surface of the substrate, and the controller may compare the first contact angle of the first data and the first contact angle of the second data, and compare the second contact angle of the first data and the second contact angle of the second data, and when the first contact angle and the second contact angle of the second data are out of error ranges of the first contact angle and the second contact angle of the first data, the controller may determine whether to load the substrate into the drying chamber and stores the second data.

(17) In the exemplary embodiment, the stored second data may be provided as analysis data when drying defect of the substrate occurs in the drying chamber.

(18) In the exemplary embodiment, the transfer robot may further include a shade member disposed above the hand, and the optical system may be provided while being installed in the shade member.

(19) In the exemplary embodiment, the optical system may measure a height of the liquid film.

(20) In the exemplary embodiment, a plurality of liquid treatment chambers may be provided while being stacked in an array of N columns and M rows, a plurality of drying chambers may be provided while being stacked in an array of n columns and m rows, and the transfer robot may transfer the substrate to a position according to a recipe between the plurality of liquid treatment chambers and the plurality of drying chambers.

(21) In the exemplary embodiment, the drying chamber may be a supercritical treatment chamber which dries the substrate with a supercritical fluid.

(22) Another exemplary embodiment of the present invention provides a method of treating a substrate by transferring the substrate from a first chamber to a second chamber, in which the transferred substrate is wetted with a chemical liquid and is transferred by a transfer robot in the state of being formed with a liquid film, the transfer robot includes a hand which is movable along an X-axis, a Y-axis, and a Z-axis and is rotatably driven based on the Z-axis, and on which the substrate is placed, and a form of the liquid film of the substrate is photographed and the photographed form of the liquid film is measured.

(23) In the exemplary embodiment, first data obtained by photographing the form of the liquid film may be stored in the state where the transfer robot picks up the substrate in the first chamber, before the transfer robot injects the substrate into the second chamber, second data obtained by photographing the form of the liquid film may be stored, and a controller may compare the first data and the second data, and when the form of the liquid film of the second data is out of an error range in comparison with the first data, the controller may determine whether to load the substrate into the second chamber and stores the second data.

(24) In the exemplary embodiment, first data obtained by photographing the form of the liquid film may be stored in the state where the transfer robot picks up the substrate in the first chamber, before the transfer robot injects the substrate into the second chamber, second data obtained by photographing the form of the liquid film may be stored, and a controller may measure a first contact angle that is a contact angle of the chemical liquid with respect to a first lateral surface of the substrate and a second contact angle that is a contact angle of the chemical liquid with respect to a second lateral surface of the substrate, and the controller may compare the first contact angle of the first data and the first contact angle of the second data, and compare the second contact angle of the first data and the second contact angle of the second data, and when the first contact angle and the second contact angle of the second data are out of error ranges of the first contact angle and the second contact angle of the first data, the controller may determine whether to load the substrate to the second chamber and stores the second data.

(25) In the exemplary embodiment, the stored second data may be provided as analysis data when drying defect of the substrate occurs in the drying chamber.

(26) In the exemplary embodiment, the first chamber may be a liquid treatment chamber, and the second chamber may be a drying chamber.

(27) In the exemplary embodiment, the drying chamber may be a supercritical treatment chamber which dries the substrate with a supercritical fluid.

(28) According to various exemplary embodiments of the present invention, it is possible to efficiently treat a substrate.

(29) According to various exemplary embodiments of the present invention, it is possible to solve drying defect that may occur due to the process in which the substrate moves between the liquid treatment chamber and the drying chamber.

(30) According to the exemplary embodiments of the present invention, the liquid treatment facility including the liquid treatment chamber, a monitoring system for the liquid film discharged and formed on the surface of the substrate W is provided, so that it is possible to solve a problem in the part unit when the problem occurs after the drying completion.

(31) The effect of the present invention is not limited to the foregoing effects, and those skilled in the art may clearly understand non-mentioned effects from the present specification and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a problem occurring in a cleaning facility in the related art.

- (2) FIG. 2 is a top plan view illustrating a substrate treating facility according to an exemplary embodiment of the present invention.
- (3) FIG. 3 is a cross-sectional view of the substrate treating facility of FIG. 2 viewed in a direction of A-A.
- (4) FIG. 4 is a top plan view schematically illustrating a transfer robot according to a first exemplary embodiment of the present invention.
- (5) FIG. 5 is a perspective view schematically illustrating a transfer robot according to a second exemplary embodiment of the present invention.
- (6) FIG. 7 is a cross-sectional view illustrating an example of a behavior of a solvent.
- (7) FIG. 6 is a front view illustrating the transfer robot of FIG. 5 viewed from the front.
- (8) FIG. 8 is a diagram illustrating a flow of a substrate treating method according to an exemplary embodiment of the present invention.
- (9) FIG. 9 is a diagram illustrating a flow of a substrate treating method according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

- (10) Hereinafter, an exemplary embodiment of the present invention will be described more fully hereinafter with reference to the accompanying drawings, in which exemplary embodiments of the invention are shown. However, the present invention can be variously implemented and is not limited to the following embodiments. In addition, in describing an exemplary embodiment of the present invention in detail, if it is determined that a detailed description of a related well-known function or configuration may unnecessarily obscure the gist of the present invention, the detailed description thereof will be omitted. In addition, the same reference numerals are used throughout the drawings for parts having similar functions and actions.
- (11) In addition, unless explicitly described to the contrary, the word “comprise” and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. It will be appreciated that terms “including” and “having” are intended to designate the existence of characteristics, numbers, steps, operations, constituent elements, and components described in the specification or a combination thereof, and do not exclude a possibility of the existence or addition of one or more other characteristics, numbers, steps, operations, constituent elements, and components, or a combination thereof in advance.
- (12) Singular expressions used herein include plurals expressions unless they have definitely opposite meanings in the context. Accordingly, shapes, sizes, and the like of the elements in the drawing may be exaggerated for clearer description.
- (13) An expression, “and/or”, includes each of the mentioned items and all of the combinations including one or more of the items. Further, in the present specification, “connected” means not only when member A and member B are directly connected, but also when member A and member B are indirectly connected by interposing member C between member A and member B.
- (14) The exemplary embodiment of the present invention may be modified in various forms, and the scope of the present invention should not be construed as being limited to the following exemplary embodiments. The present exemplary embodiment is provided to more completely explain the present invention to those skilled in the art. Therefore, the shapes of elements in the drawings are exaggerated to emphasize clearer descriptions.
- (15) FIG. 2 is a top plan view illustrating a substrate treating facility according to an exemplary embodiment of the present invention. FIG. 3 is a cross-sectional view of the substrate treating facility of FIG. 2 viewed in a direction of A-A. The substrate treating facility will be described with reference to FIGS. 2 and 3. The substrate treating facility 1 treating a substrate includes an index module 10 and a process processing module 20. The index module 10 includes a load port 120 and a transfer frame 140. The load port 120, the transfer frame 140, and the process processing module 20 may be sequentially arranged in series. Hereinafter, a direction in which the load port 120, the

transfer frame **140**, and the process processing module **20** are arranged is called to as a first direction **12**, and a direction perpendicular to the first direction **12** when viewed from the top is called a second direction **14**, and a direction perpendicular to a plane including the first direction **12** and the second direction **14** is called a third direction **16**.

(16) A carrier **18** in which a substrate W is accommodated is seated on the load port **120**. The load port **120** is provided in plurality, and the plurality of load ports **120** is arranged in series in the second direction **14**. In FIG. 2, it is illustrated that four load ports **120** are provided. However, the number of load ports **120** may be increased or decreased according to process efficiency of the process processing module **20** and a condition of foot print, and the like. Slots (not illustrated) provided to support an edge of the substrate are formed in the carrier **18**. A plurality of slots is provided in the third direction **16**, and the substrates are positioned within the carrier so as to be stacked while being spaced apart from each other in the third direction **16**. As the carrier **18**, a Front Opening Unified Pod (FOUP) may be used.

(17) The process processing module **20** includes a buffer unit **220**, a transfer chamber **240**, and a liquid treatment chamber **260** and a drying chamber **280** as treatment chambers. The transfer chamber **240** is disposed so that a longitudinal direction thereof is parallel to the first direction **12**. In the liquid treatment chamber **260** and the drying chamber **280** as the treatment chambers, the substrate is treated therein, and an entrance through which the substrate enters and exits is formed. The liquid treatment chambers **260** are disposed at one side of the transfer chamber **240** in the second direction **14**, and the drying chambers **280** are disposed at the other side of the transfer chamber **240**. The liquid treatment chambers **260** and the drying chambers **280** may be provided to be symmetric to each other based on the transfer chamber **240**. Some of the liquid treatment chambers **260** are disposed in the longitudinal direction of the transfer chamber **240**. Further, some of the liquid treatment chambers **260** are disposed to be stacked with each other. That is, the liquid treatment chambers **260** may be disposed in an array of $A \times B$ (each of A and B is a natural number equal to or larger than 1) at one side of the transfer chamber **240**. Herein, A is the number of liquid treatment chambers **260** provided in series in the first direction **12**, and B is the number of liquid treatment chambers **260** provided in series in the third direction **16**. When six or nine liquid treatment chambers **260** are provided at one side of the transfer chamber **240**, the liquid treatment chambers **260** may be disposed in an array of 3×2 or 3×3 . The number of liquid treatment chambers **260** may also increase or decrease. Similar to the liquid treatment chambers **260**, the drying chambers **280** may be arranged in an array of $M \times N$ (each of M and N is a natural number equal to or larger than 1). Herein, M and N may be the same as A and B, respectively. Unlike the foregoing description, both the liquid treatment chambers **260** and the drying chambers **280** may be provided only at one side of the transfer chamber **240**. Further, unlike the foregoing description, each of the liquid treatment chambers **260** and the drying chamber **280** may be provided in a single layer at one side and the other side of the transfer chamber **240**. Optionally, the liquid treatment chamber **260**s may be positioned to be stacked at one side of the transfer chamber **240**, and the drying chambers **280** may be positioned to be stacked at the other side of the transfer chamber **240**. Further, the liquid treatment chamber **260**s and the drying chambers **280** may be provided in various dispositions unlike the foregoing description.

(18) The buffer unit **220** is disposed between the transfer frame **140** and the transfer chamber **240**. The buffer unit **220** provides a space in which the substrate W stays before the substrate W is transferred between the transfer chamber **240** and the transfer frame **140**. Slots (not illustrated) on which the substrate W is placed are provided inside the buffer unit **220**, and the plurality of slots (not illustrated) is provided so as to be spaced apart from each other in the third direction **16**. A surface of the buffer unit **220** facing the transfer frame **140** and a surface of the buffer unit **220** facing the transfer chamber **240** are opened.

(19) The transfer frame **140** transfers the substrate W between the carrier **18** seated on the load port **120** and the buffer unit **220**. An index rail **142** and an index robot **144** are provided to the transfer

frame **140**. The index rail **142** is provided so that a longitudinal direction thereof is parallel to the second direction **14**. The index robot **144** is installed on the index rail **142**, and linearly moves in the second direction **14** along the index rail **142**. The index robot **144** includes a base **144a**, a body **144b**, and an index arm **144c**. The base **144a** is installed to be movable along the index rail **142**. The body **144b** is coupled to the base **144a**. The body **144b** is provided to be movable in the third direction **16** on the base **144a**. Further, the body **144b** is provided to be rotatable on the base **144a**. The index arm **144c** is coupled to the body **144b** and is provided to be movable forwardly and backwardly with respect to the body **144b**. A plurality of index arms **144c** is provided to be individually driven. The index arms **144c** are disposed to be stacked in the state of being spaced apart from each other in the third direction **16**. A part of the index arms **144c** may be used when the substrate W is transferred from the process processing module **20** to the carrier **18**, and another part of the plurality of index arms **144c** may be used when the substrate W is transferred from the carrier **18** to the process processing module **20**. This may prevent the particles generated from the substrate W before the process processing from being attached to the substrate W after the process processing in the process in which the index robot **144** loads and unloads the substrate W.

(20) The transfer robot **244** transferring the substrate W into and out of the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers through the entrances of the liquid treatment chamber **260** and the drying chamber **280** is provided to the transfer chamber **240**. According to the exemplary embodiment, the transfer robot **244** transfers the substrate W between the buffer unit **220**, the liquid treatment chamber **260**, and the drying chamber **280**. The transfer robot **244** is provided to be movable along the guide rail **242**. The guide rail **242** is disposed so that the longitudinal direction thereof is parallel to the first direction **12**. The transfer robot **244** is installed on the guide rail **242**, and linearly moves on the guide rail **242** in the first direction **12**. Further, the transfer robot **244** is movable up and down in the third direction **16**. Further, the transfer robot **244** is rotatable in the third direction as an axis. In the transfer chamber **240**, a fan unit **246** generating a downdraft in a transferring region therein is provided. The downdraft prevents inside foreign substances, such as particles and fumes, from floating. When the substrate W is transferred from the liquid treatment chamber **260** to the drying chamber **280**, the substrate W is transferred in the state where the liquid supplied from the liquid treatment chamber **260** remains.

(21) FIG. **4** is a top plan view schematically illustrating the transfer robot according to a first exemplary embodiment of the present invention. The transfer robot will be described with reference to FIG. **4**. A transfer robot **2441** according to a first exemplary embodiment of the present invention is an example of the transfer robot **244**. The transfer robot **2441** includes a base **710** and a hand **720**.

(22) The base **710** is linearly movable in a horizontal direction and a vertical direction, and may be provided to be rotatable based on an axis parallel to the vertical direction. The hand **720** is coupled to the base **710**. According to the exemplary embodiment, the hand **720** is provided above the base **710**.

(23) The hand **720** grips the substrate W. The hand **720** is provided to be linearly movable forward and backward. For example, the hand **720** moves forward from the top of the base **710** to take over the substrate W at a position where a seating portion **721** of the hand **720** deviates from an upper surface of the base **710**. Thereafter, the transfer robot **244** moves the base **710** in the state where the hand **720** moves backward in the state where the substrate W is placed and is located at a position facing the upper surface of the base **710**, to move the base **710** to a position adjacent to a target position at which the substrate W is to be placed. Thereafter, the hand **720** takes over the substrate W to the target position in the state of being moved forward so as to be placed above the target position.

(24) According to the exemplary embodiment, the hand **720** includes the seating part **721** and guide protrusions **725**. The substrate W is seated on the seating part **721**. A plurality of protrusions

supporting the bottom surface of the substrate W may be provided to an upper surface of the seating part **721**. The guide protrusion **725** supports a front lateral portion of the substrate W that is seated in the proper position of the seating part **721**. The guide protrusion **725** protrudes upward from the seating part **721**. The plurality of guide protrusions **725** may be provided to be spaced apart from each other.

(25) A driving member (not illustrated) moves the hand **720** forward and backward. According to the exemplary embodiment, the driving member (not illustrated) includes a motor providing driving force moving the hand **720**.

(26) According to the exemplary embodiment, a camera **910** is provided to the transfer robot **244**. The camera **910** includes a first camera **911** photographing a first side and a second camera **912** photographing a second side. The camera **910** photographs the substrate W in the horizontal direction and monitors an edge region of a liquid film L. Data captured by the camera **910** is transmitted to a controller **920**. The controller **920** measures a contact angle of the liquid film L. As a more particular example, the controller **920** measures a contact angle of the edge region of the liquid film L. The case of utilizing the measured contact angle will be described below in detail. The camera **910** may be positioned above a pusher guide (not illustrated) for adjusting the position of the substrate W placed on the seating part **721** to the proper position.

(27) FIG. 5 is a perspective view schematically illustrating a transfer robot according to a second exemplary embodiment of the present invention. FIG. 6 is a front view illustrating the transfer robot of FIG. 5 viewed from the front. A transfer robot **2442** will be described with reference to FIGS. 5 and 6 as an example of the transfer robot **244** provided in FIGS. 2 and 3. The transfer robot **2442** includes a base **710**, a hand **720**, and a shade member **730**.

(28) The base **710** is provided to be moved along the guide rail **242**. The base **710** supports the hand **720**.

(29) The substrate W is placed on the hand **720**. The hand **720** is provided to be stretchable relative to the base **710**. The hand **710** directly loads or unloads the substrate W into or from the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers through the entrances. The plurality of hands **720** may be provided to be stacked with each other.

(30) The shade member **730** minimizes a downdraft formed by the fan unit **246** from reaching the substrate placed on the hand **720**. The shade member **730** is fixedly coupled to the base **710** so as to be disposed above the hand **720**. When the hand **720** loads or unloads the substrate W into or from the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers, the shade member **730** may be disposed at a position higher than the upper ends of entrances **261** and **281**. When viewed from the top, the shade member **730** is provided with a larger area than that of the substrate W.

(31) The shade member **730** is provided inclined so that a height is lowered from the center toward both sides. Therefore, the downdraft flows out of the region of the hand **720** where the substrate W is provided along the inclined surface of the shade member **730**. When viewed from the top, the shade member **730** may be provided so that one lateral surfaces of two plates are fixedly coupled to each other in the central region of the hand **720**. Each plate may be provided to be inclined in a down direction from the center region to the edge region of the hand **720**. When the hand **720** loads or unloads the substrate into or from the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers, the shade member **730** is disposed at a position higher than the upper ends of the entrances of the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers. The shade member **730** is provided to be sufficiently adjacent to the entrances **261** and **281** so as to prevent the downdraft from being introduced through the entrances **261** and **281**. As described above, the shade member **730** is provided above the hand **720**, so that the downdraft is prevented from being in directly contact to the substrate W placed on the hand **720** during the transferring of the substrate within the transfer chamber **240**.

(32) Further, the transfer robot **2442** may suppress the downdraft from being introduced through

the entrances when the substrate W is loaded/unloaded into/from the liquid treatment chamber **260** and the drying chamber **280** that are the treatment chambers. The shade member **730** is provided with an uninterruptible material for preventing the generation of static electricity. For example, the shade member **730** may be made of an uninterruptible vinyl chloride resin (PVC).

(33) According to the exemplary embodiment, a camera **1910** is installed in the shade member **730**. The camera **1910** may be installed at the highest position of the shade member **730**. The camera **1910** may be provided as a camera capable of measuring a height of the liquid film. The camera **1910** photographs the substrate W from the above and monitors the height of the liquid film L. Data captured by the camera **1910** is transmitted to a controller **920**. The controller **1910** calculates a height for each region of the liquid film L. As a more particular example, the controller **920** calculates a height of the edge region of the liquid film L. The case of utilizing the measured height will be described below in detail.

(34) FIG. 7 is a cross-sectional view illustrating an example of a behavior of a solvent. A behavior of a solvent L as the liquid film formed on the substrate W will be described with reference to FIG. 7. For example, even though the amount of chemical liquid discharged to the surface of the substrate W is set to 10 g at the time of initial set-up and the amount of chemical liquids is the same according to the behavior of the chemical liquid while the substrate W moves to the drying chamber **280**, a thickness of the chemical liquid L may be different for each region on the surface of the substrate W. For example, as referenced from the drawings of (b), (c), and (d), the case where the liquid film L is raised to the left because the chemical liquid is biased to the left ((b) the example of the left rise), the case where the liquid film L is raised to the right because the chemical liquid is biased to the right ((c) the example of the right rise), and the case where the liquid film L is raised because the chemical liquid is collected to the center ((d) the example of the center rise) are the representative cases. Like the representative example, when the behavior of the chemical liquid occurs and the substrate is moved to the drying chamber **280** in the non-uniform state of the liquid film L and then drying proceeds, since the evaporation amount is different depending on the thickness of the chemical liquid L, local evaporation may occur first. The local evaporation eventually leads to pattern collapse or leaning which are drying defects.

(35) FIG. 8 is a diagram illustrating a flow of a substrate treating method according to an exemplary embodiment of the present invention. FIG. 9 is a diagram illustrating a flow of a substrate treating method according to an exemplary embodiment of the present invention. The substrate treating method will be described with reference to FIGS. 8 and 9.

(36) After a substrate W is wetted with a chemical liquid (for example, an IPA) in the liquid treatment chamber **260**, a form of a liquid film is measured in the state where the transfer robot **244** picks up the substrate W and the measured form of the liquid film is stored as first data. Then, the form of the liquid film is measured before the substrate W is injected into the drying chamber **280** and the measured form of the liquid film is stored as second data. The first data and the second data are compared. As described in FIG. 8, when the form of the liquid film according to the second data and the form of the liquid film according to the first data are compared and a result of the comparison is out of a set error range, whether to load the substrate W to the drying chamber **280** is determined and the data is stored. When the drying defect occurs in the substrate W, the stored data may be used for post analysis. As described in FIG. 9, when the form of the liquid film according to the second data and the form of the liquid film according to the first data are compared and a result of the comparison is within the set error range, the substrate W is loaded into the drying chamber **280** and the data is stored. When the drying defect occurs in the substrate W, the stored data may be used for post analysis.

(37) According to the first exemplary embodiment, the measurement of the form of the chemical liquid is to measure a contact angle of the chemical liquid. After the substrate W is wetted with a chemical liquid (for example, an IPA) in the liquid treatment chamber **260**, the camera **910** photographs an edge region of the liquid film L in the state where the transfer robot **244** picks up

the substrate W. The controller **920** measures the contact angle of the liquid film L from the photographed data and stores the measured contact angle as first data. Then, the camera **910** photographs the edge region of the liquid film L before the substrate W is injected into the drying chamber **280**. The controller **920** measures the contact angle of the liquid film L and stores the measured contact angle as second data. The controller **920** compares the first data and the second data. When the contact angle of the liquid film according to the second data is compared with the contact angle of the liquid film according to the first data and a result of the comparison is out of a set error range, whether to load the substrate W to the drying chamber **280** is determined and the data is stored. For example, when the contact angle of the liquid film according to the second data is compared with the contact angle of the liquid film according to the first data and the result of the comparison is out of the set error range, the controller **920** waits for a set time until the liquid film L on the substrate W stabilizes itself, and when a result of the re-measurement is within the error range, the substrate W may be loaded into the drying chamber **280**.

(38) According to the second exemplary embodiment, the measurement of the form of the chemical liquid is to measure a height of the chemical liquid for each region. After the substrate W is wetted with a chemical liquid (for example, an IPA) in the liquid treatment chamber **260**, the camera **1910** photographs an entire region of the liquid film L in the state where the transfer robot **244** picks up the substrate W. A controller **1920** measures a height of the liquid film L for each region from the photographed data and stores the measured height as first data. Then, the camera **1910** photographs the entire region of the liquid film L before the substrate W is injected into the drying chamber **280**. The controller **1920** measures the height of the liquid film L for each region and stores the measured height as second data. The controller **1920** compares the first data and the second data. When the height of the liquid film for each region according to the second data is compared with the height of the liquid film for each region according to the first data and a result of the comparison is out of a set error range, or the height of the liquid film for each region according to the second data is such that the pattern collapse is likely to occur, whether to load the substrate W to the drying chamber **280** is determined and the data is stored. For example, when the height of the liquid film according to the second data is compared with the height of the liquid film according to the first data and the result of the comparison is out of the set error range, the controller **1920** waits for a set time until the liquid film L on the substrate W stabilizes itself, and when a result of the re-measurement is within the error range, the substrate W may be loaded into the drying chamber **280**.

(39) As the exemplary embodiment of the present invention, the camera **910** is provided to the hand of the transfer robot **244**, but the camera **910** may be provided to an end of a bowl that collects the treatment liquid in the cleaning chamber. Further, the camera **910** may also be provided to both sides of the entrance through which the substrate W is inserted in the drying chamber **280**. If the camera **910** is capable of measuring the contact angle of the edge region of the substrate W as in the first embodiment of the present invention, the design may be changed so that the camera **910** is provided in other places other than the presented example.

(40) According to the exemplary embodiment of the present invention, the liquid treatment device and facility, the monitoring system for the liquid film discharged and formed on the surface of the substrate W is provided, so that it is possible to solve a problem in the part unit when the problem occurs after the drying completion.

(41) The controller may control the constituent elements of the substrate treating apparatus and facility so that the substrate is treated according to a set process. Further, the controller may include a process controller formed of a microprocessor (computer) executing the control of the substrate treating apparatus and facility, a keyboard on which a command input manipulation is performed in order for an operator to manage the substrate treating apparatus and facility, a user interface formed of a display and the like visualizing and displaying an operation situation of the substrate treating apparatus, and a storage unit in which a control program for executing the process executed in the substrate treating apparatus and facility under the control of the controller or a program, that is, a

processing recipe, for executing the processing to each configuration according to various data and processing conditions, are stored. Further, the user interface and the storage unit may be connected to the process controller. The processing recipe may be stored in a storage medium in the storage unit, and the storage medium may be a hard disk, and may also be a portable disk, such as a CD-ROM or a DVD, or a semiconductor memory, such as a flash memory.

(42) The foregoing detailed description illustrates the present invention. Further, the above content shows and describes the exemplary embodiment of the present invention, and the present invention can be used in various other combinations, modifications, and environments. That is, the foregoing content may be modified or corrected within the scope of the concept of the invention disclosed in the present specification, the scope equivalent to that of the disclosure, and/or the scope of the skill or knowledge in the art. The foregoing exemplary embodiment describes the best state for implementing the technical spirit of the present invention, and various changes required in specific application fields and uses of the present invention are possible. Accordingly, the detailed description of the invention above is not intended to limit the invention to the disclosed exemplary embodiment. Further, the accompanying claims should be construed to include other exemplary embodiments as well.

Claims

1. A substrate treating apparatus, comprising: a liquid treatment chamber configured to treat a substrate with a liquid; a drying chamber configured to dry the liquid-treated substrate; a transfer robot configured to transfer the substrate between the liquid treatment chamber and the drying chamber, and including a hand which is movable along an X-axis, a Y-axis, and a Z-axis and is rotatably driven based on the Z-axis, and on which the substrate is placed; an optical system configured to photograph a form of a liquid film of the substrate, in which when the substrate is transferred from the liquid treatment chamber to the drying chamber, the substrate is wetted with a chemical liquid and is transferred by the transfer robot in a state of being formed with & the liquid film formed; and a controller configured to measure the form of the liquid film photographed by the optical system, the form including a contact angle of the liquid film and the substrate.
2. The substrate treating apparatus of claim 1, wherein the optical system transmits first data obtained by photographing the form of the liquid film to the controller in the state where the transfer robot picks up the substrate in the liquid treatment chamber, before the transfer robot injects the substrate into the drying chamber, the optical system transmits second data obtained by photographing the form of the liquid film to the controller, and the controller compares the first data and the second data, and when the form of the liquid film of the second data is out of an error range in comparison with the first data, the controller determines whether to load the substrate into the drying chamber and stores the second data.
3. The substrate treating apparatus of claim 2 wherein the stored second data is provided as analysis data when drying defect of the substrate occurs in the drying chamber.
4. The substrate treating apparatus of claim 3, wherein the optical system measures a height of the liquid film.
5. The substrate treating apparatus of claim 1, wherein the optical system is a camera which photographs an edge region of the substrate from a lateral surface of the substrate.
6. The substrate treating apparatus of claim 5, wherein the camera is provided to photograph a first lateral surface of the substrate and a second lateral surface that is the other side of the first lateral surface.
7. The substrate treating apparatus of claim 6, wherein the controller measures a first contact angle that is a contact angle of the chemical liquid with respect to the first lateral surface of the substrate and a second contact angle that is a contact angle of the chemical liquid with respect to the second lateral surface of the substrate.

8. The substrate treating apparatus of claim 1, wherein the optical system transmits first data obtained by photographing the form of the liquid film to the controller in the state where the transfer robot picks up the substrate in the liquid treatment chamber, before the transfer robot injects the substrate into the drying chamber, the optical system transmits second data obtained by photographing the form of the liquid film to the controller, the controller measures a first contact angle that is a contact angle of the chemical liquid with respect to a first lateral surface of the substrate and a second contact angle that is a contact angle of the chemical liquid with respect to a second lateral surface of the substrate, and the controller compares the first contact angle of the first data and the first contact angle of the second data, and compares the second contact angle of the first data and the second contact angle of the second data, and when the first contact angle and the second contact angle of the second data are out of error ranges of the first contact angle and the second contact angle of the first data, the controller determines whether to load the substrate into the drying chamber and stores the second data.

9. The substrate treating apparatus of claim 1, wherein the transfer robot further includes a shade member disposed above the hand, and the optical system is provided while being installed in the shade member.

10. The substrate treating apparatus of claim 1, wherein a plurality of liquid treatment chambers is provided while being stacked in an array of N columns and M rows, a plurality of drying chambers is provided while being stacked in an array of n columns and m rows, and the transfer robot transfers the substrate to a position according to a recipe between the plurality of liquid treatment chambers and the plurality of drying chambers.

11. The substrate treating apparatus of claim 1, wherein the drying chamber is a supercritical treatment chamber which dries the substrate with a supercritical fluid.

12. A substrate treating apparatus, comprising: a plurality of liquid treatment chambers provided while being stacked in an array of N columns and M rows and configured to treat a substrate with a liquid; a plurality of drying chambers provided while being stacked in an array of n columns and m rows and configured to dry the liquid-treated substrate; a transfer robot configured to transfer the substrate between one of the liquid treatment chambers and one of the drying chambers, and including a hand which is movable along an X-axis, a Y-axis, and a Z-axis and is rotatably driven based on the Z-axis, and on which the substrate is placed; an optical system configured to photograph a form of a liquid film of the substrate, in which when the substrate is transferred from the one of the liquid treatment chambers to the one of the drying chambers, the substrate is wetted with a chemical liquid and is transferred by the transfer robot in a state of being formed with the liquid film; and a controller configured to measure the form of the liquid film photographed by the optical system, wherein the optical system transmits first data obtained by photographing the form of the liquid film to the controller in a state where the transfer robot picks up the substrate in the one of the liquid treatment chambers, before the transfer robot injects the substrate into the one of the drying chambers, the optical system transmits second data obtained by photographing the form of the liquid film to the controller, the controller compares the first data and the second data, and when the form of the liquid film of the second data is out of an error range in comparison with the first data, the controller determines whether to load the substrate into the one of the drying chambers and stores the second data, and the stored second data is provided as analysis data when drying defect of the substrate occurs in the one of the drying chambers.
